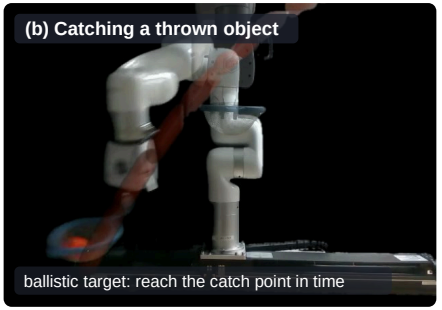
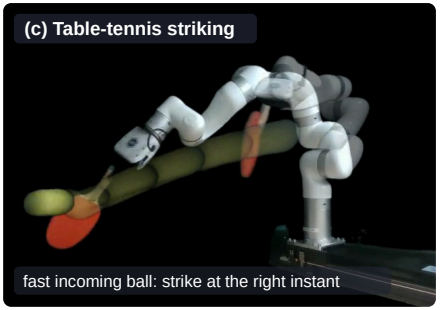
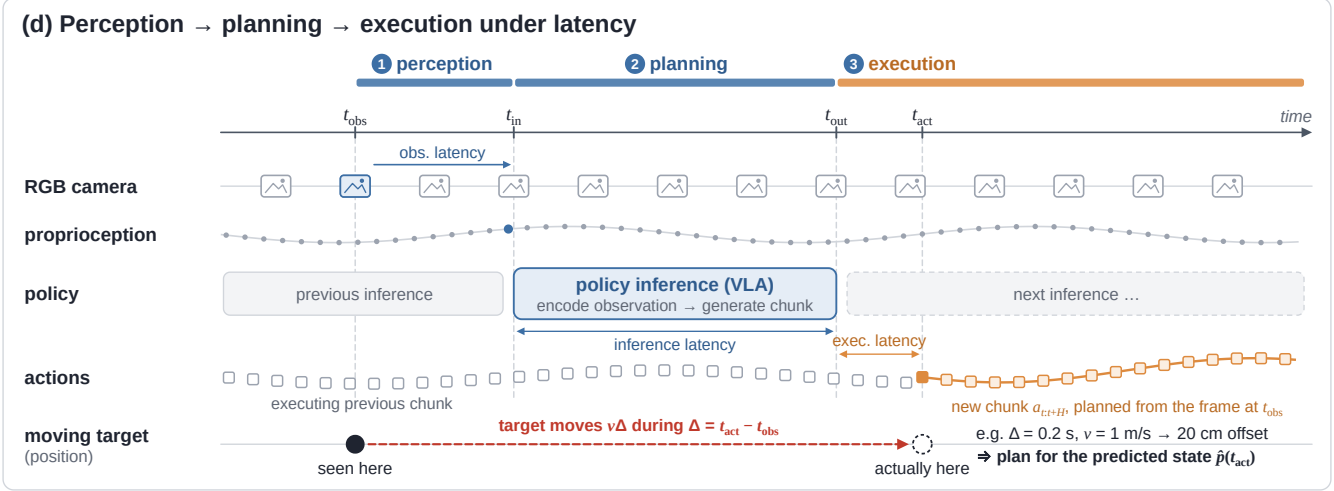

(a) Conveyor sorting
target keeps moving: grasp on the move


(b) Catching a thrown object
ballistic target: reach the catch point in time


(c) Table-tennis striking
fast incoming ball: strike at the right instant


(d) Perception → planning → execution under latency
1 perception
2 planning
3 execution
t_{obs}
t_{in}
t_{out}
t_{act}
time
obs. latency
RGB camera
proprioception
policy
previous inference
policy inference (VLA)
encode observation → generate chunk
next inference …
inference latency
exec. latency
actions
executing previous chunk
new chunk $a_{t:t+H}$, planned from the frame at t_{obs}
moving target
(position)
target moves $v\Delta$ during $\Delta = t_{act} - t_{obs}$
e.g. Δ = 0.2 s, v = 1 m/s → 20 cm offset
seen here
actually here
⇒ plan for the predicted state $\hat{p}(t_{act})$